Math Inquirers

Extension Workbook

with answers

Level 6

Self-study | Ages 11-13



My name _____

About this book

Math Inquirers Extension Workbook, Level 6
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

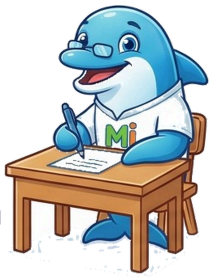
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 23: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- **Start here.** The main path. Do these first.
- **Think harder.** Longer problems. Hints are printed upside down.
- **Go further.** A weekend page. Big challenges, just for fun.
- **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	**Code the pattern**	6-7	8-9	10 and 12	11	☐
2	**Prime hunters**	13-14	15-16	17 and 19	18	☐
3	**Below zero**	Not in this proof edition				☐
4	**Algebra machines**	Not in this proof edition				☐
5	**Every possible way**	Not in this proof edition				☐
6	**Fair or unfair?**	Not in this proof edition				☐
7	**Turtle geometry**	Not in this proof edition				☐
8	**Secret codes**	Not in this proof edition				☐
9	**Data detectives**	Not in this proof edition				☐
10	**Showcase challenge**	Not in this proof edition				☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

Learn it: Sequences and the nth term

My goal. I can find the rule for the nth term of a pattern and use it for the 100th term. I can test my rule with a loop.

A rule for the nth term jumps straight to any term, such as the 100th. The difference tells you which times table to compare with. A loop in code adds the difference again and again, so you can test your rule.

Worked example

A sequence begins 9, 14, 19, 24, … Find the rule that gives term n. Then find the 100th term.

n	1	2	3	4
5 × n	5	10	15	20
term	9	14	19	24

1. **The difference is 5, so compare each term with the 5 times table.**
2. **Each term is 4 more than the number above it. The rule is 5n + 4.**
3. **Check the 4th term: 5 × 4 + 4 = 24. It matches.**
4. **The 100th term is 5 × 100 + 4 = 504.**
5. **Test it in code: set term to 9, then repeat 99 times: change term by 5. The 100th term is 99 jumps after the 1st term.**

Answer: The rule is 5n + 4. The 100th term is 504.

How to find the rule for the nth term

1. Find the difference between neighbouring terms.
2. Write that times table under n: 5, 10, 15, …
3. Find how much to add or take away to reach each term.
4. Check the rule on 2 terms. Then use it for any term.

Your turn

Answer at the back of the book.

Try

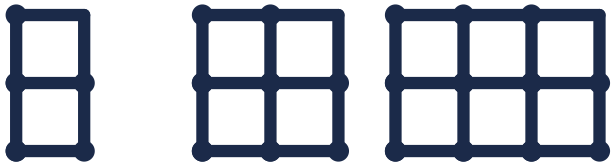
A sequence starts 3, 10, 17, 24, …

The rule for the nth term is ☐ × n − ☐. The 100th term is ☐.

Start here

Do these first. Take your time.

1 Matchsticks make 2 rows of squares. Pattern 2 has 2 columns and pattern 3 has 3 columns.



Pattern 1 Pattern 2 Pattern 3

(a) Pattern 3 has ☐ sticks. Each new column adds ☐ sticks.

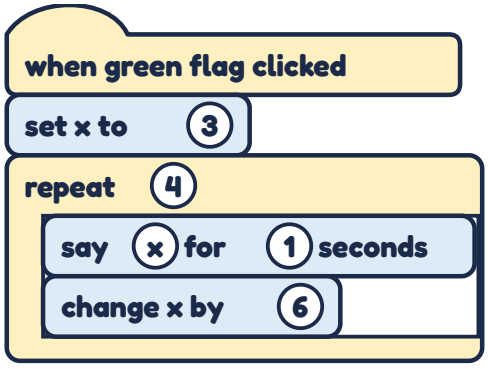
(b) The rule: sticks = ☐ × n + ☐

(c) Pattern 40 has ☐ sticks. (d) Pattern ☐ has 92 sticks.

2 Complete the table for each sequence.

Sequence	Difference	Rule for term n	Term 100
(a) 8, 14, 20, 26, …			
(b) 3, 8, 13, 18, …			

3 Trace the code. The sprite says x first, then changes it. Fill in the trace table.



Loop	x said	x after
1		
2		
3		
4		

(a) The last number it says is ☐.

(b) With repeat 50, the last number it says is ☐.

Start here

Keep going. Check your work.

4 **A sequence has the rule 5n + 2. Its first terms are 7, 12, 17, 22, ...**

(a) Is 157 a term? Yes No If yes, it is term number ☐.

(b) Is 205 a term? Yes No

(c) The 60th term is ☐.

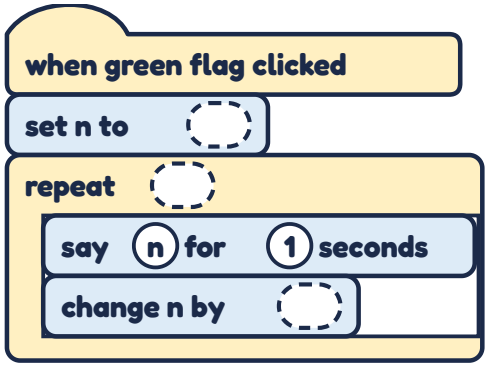
5 **Complete the code so that the sprite says the first 15 terms of 11, 18, 25, 32, ...**

(a) set n to ☐

(b) repeat ☐

(c) change n by ☐

(d) The last number it says is ☐.

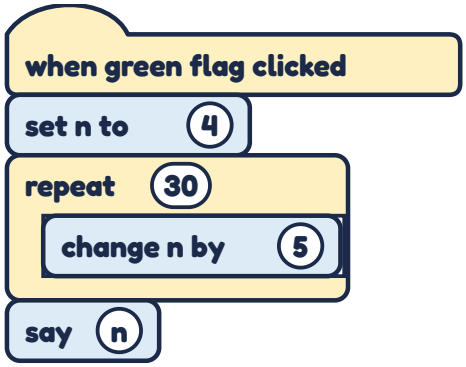


6 **Siti's code should say the 30th term of 4, 9, 14, 19, ...**

(a) Her code says ☐.

(b) The 30th term really is ☐.

(c) To fix it, change repeat 30 to repeat ☐.



7 **Circle the rule that gives each sequence. Test n = 1 and n = 2.**

(a) 4, 9, 14, 19, ... 4n + 1 5n – 1 5n + 4

(b) 10, 13, 16, 19, ... 3n + 10 10n + 3 3n + 7

(c) 3, 11, 19, 27, ... 8n + 3 8n – 5 3n + 8

Start here

Nearly there. You can do it.

8 **Ali has $15 on Day 1. Every day after that he adds $7. So on Day 2 he has $22.**

(a) On Day 4 he has $ ☐ .

(b) Money on Day n = ☐ × n + ☐ dollars.

(c) On Day 30 he has $ ☐ .

(d) He first has more than $200 on Day ☐ .

9 **A program begins on the 1st term of a sequence. Each repeat moves it on to the next term.**

(a) To reach the 60th term, repeat ☐ times.

(b) After repeat 40, the sprite is at term number ☐ .

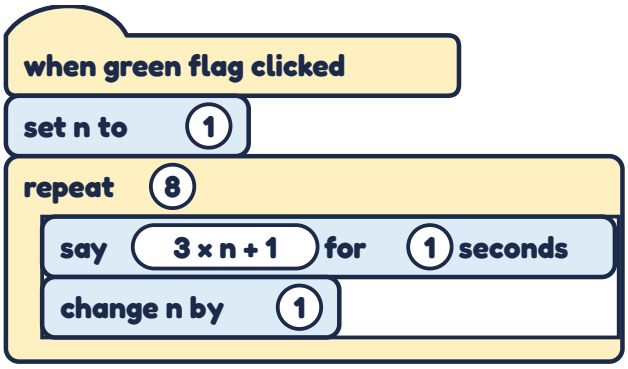
(c) To go from the 8th term to the 25th term, repeat ☐ times.

10 **This code uses n as a counter and says a rule. Predict what it says.**

(a) It starts: _____

(b) The last number it says is ☐ .

(c) To say 5, 9, 13, … use ☐ × n + ☐



11 **Aisha looks at 2, 8, 14, 20, … The difference is 6, so she says the rule is 6n.**

(a) Is she right? Yes No

(b) The rule 6n gives ☐ for the 1st term, but the 1st term is ☐ .

(c) The correct rule is ☐ × n – ☐ .

Think harder

Hints are upside down. Try first!

1 **A sequence goes down by the same amount each time: 58, 53, 48, 43, …**

(a) The rule for the nth term is ☐ – 5n. (b) The 10th term is ☐.

(c) The last term above 0 is term number ☐.

Working: _____

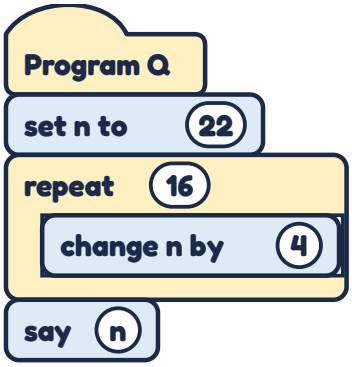
Hint. The terms go down by 5. Take 5 × n away from a start number.

2 **Program P starts at the 1st term of 6, 10, 14, 18, … Program Q starts at the 5th term of the same sequence. Both change by 4 each repeat.**

(a) Q ends at term number ☐.

(b) To end at the same term, P must repeat ☐ times.

(c) Both programs say ☐.



Hint. Count how many terms Q moves on. P starts 4 terms earlier.

3 **A sequence goes up by the same amount each time. Its 4th term is 23 and its 9th term is 53.**

(a) From the 4th term to the 9th term is ☐ jumps.

(b) Each jump is ☐. (c) The 1st term is ☐.

(d) The rule is ☐ × n – ☐.

Working: _____

Hint. Count the jumps between the 2 terms. Share the rise equally.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 Beyond Level 6: Second differences.

Matchsticks make staircases. Look at staircases 1, 2 and 3.



Staircase 1 Staircase 2 Staircase 3

(a) Count the sticks. Staircase 2 has ☐ and staircase 3 has ☐.

(b) The extra sticks go up by the same amount each time. Staircase 4 has ☐ sticks and staircase 5 has ☐.

(c) Staircase 10 has ☐ sticks.

Hint. Find how many extra sticks each staircase adds. Look for a pattern.

2 A sprite says these numbers in order: 3, 20, 7, 25, 11, 30, 15, 35, ...

(a) Places 1, 3, 5, ... go up by ☐. Places 2, 4, 6, ... go up by ☐.

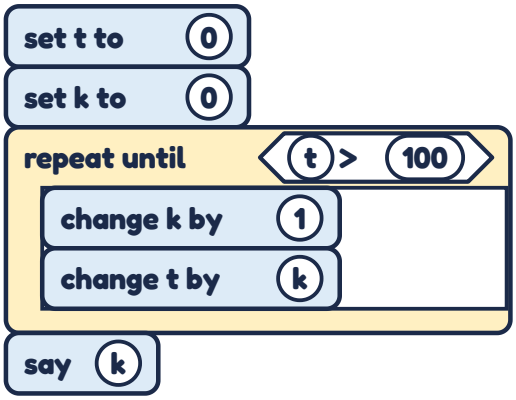
(b) The number in place 15 is ☐.

(c) The number in place 20 is ☐.

Hint. Split it into 2 sequences. Which one is place 15 in?

3 Beyond Level 6: Triangle numbers.

What does this code say? Write t after each repeat.



(a) The code says ☐.

(b) When it stops, t is ☐.

Hint. Write t after every repeat: 1, 3, 6, ... Where does it pass 100?

Name _____

✔ Check yourself

No hints. Try each one on your own.

1 Find the rule for 9, 16, 23, 30, … The rule is ☐ × n + ☐. The 100th term is ☐.

Working: _____

2 Is 125 a term of 7, 12, 17, 22, …? Yes No

Working: _____

3 The code is: set n to 8, then repeat 20 times: change n by 6. At the end n = ☐, which is term number ☐ of 8, 14, 20, …

Working: _____

4 Mei starts at n = 5. She repeats 4 times: change n by 3, then say n. What is the first number it says? ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Primes and factors

My goal. I can find all the factors of a number, test whether it is prime and write it as a product of primes. I can count factors with a loop.

A factor divides a number exactly. Factors come in pairs, and a prime has exactly 2 factors: 1 and itself. In code, n mod d is the remainder, and a remainder of 0 means d is a factor.

Worked example

Is 97 prime? Show your test.

d	2	3	5	7
97 mod d	1	1	2	6

1. **A prime has only 2 factors. We look for any factor besides 1 and 97.**
2. **Test the primes 2, 3, 5 and 7. Stop at 7, because 11 × 11 = 121 is already more than 97.**
3. **Find the remainder 97 mod d for each. None of them is 0.**
4. **A program that counts the factors of 97 ends with count = 2.**

Answer: 97 is prime: it has no factor except 1 and 97.

How to test whether n is prime

1. Try the primes 2, 3, 5, 7, … in order.
2. Find n mod d. A remainder of 0 means d is a factor.
3. Stop when d × d is more than n.
4. No remainder was 0? Then n is prime.

Your turn

Answer at the back of the book.

Try **Is 143 prime?** Yes No **If not, 143 =** ☐ × ☐ .

Start here

Do these first. Take your time.

1 **Factors come in pairs. Use pairs to find all the factors.**

(a) The factors of 42 are _____

(b) 100 has ☐ factors.

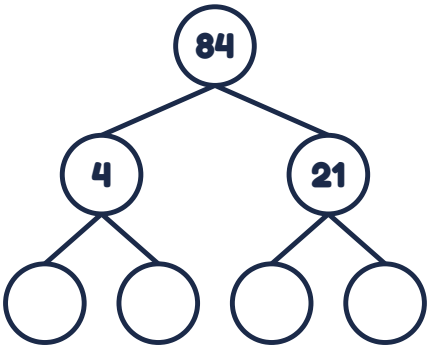
(c) The factor of 100 that pairs with itself is ☐.

2 **Circle every prime number in the table.**

51	53	57	59	63
67	69	73	77	79

3

Finish the factor tree of 84. Split each number until only primes are left.



(a) 84 = ☐ × ☐ × ☐ × ☐

(b) Write 90 as a product of primes:

_____

4 **Use the primes to write 26 as a sum of 2 primes. Find every way.**

2	3	5	7	11	13	17	19	23	29

_____

Start here

Keep going. Check your work.

5 **Use the divisibility tests. Circle Yes or No.**

(a) Is 372 a multiple of 4? Yes No

(b) Is 2457 a multiple of 3? Yes No

(c) Is 6835 a multiple of 9? Yes No

6 **Test whether 113 is prime. You only need to test 2, 3, 5 and 7.**

(a) 113 mod 2 = ☐, mod 3 = ☐, mod 5 = ☐, mod 7 = ☐

(b) Is 113 prime? Yes No

(c) Is 119 prime? Yes No If not, 119 = ☐ × ☐.

7 **Who am I? Each riddle has only one answer.**

(a) I am a prime number between 40 and 60. My digits add to 8. I am ☐.

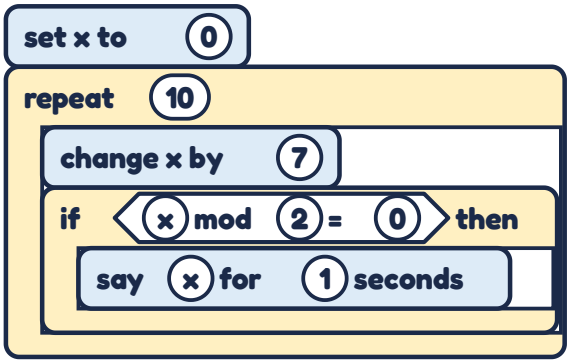
(b) I am a prime number between 60 and 80. If you add 2 to me, you get another prime. I am ☐.

8 **Predict what this code says.**

(a) The first number it says is ☐.

(b) It says ☐ numbers.

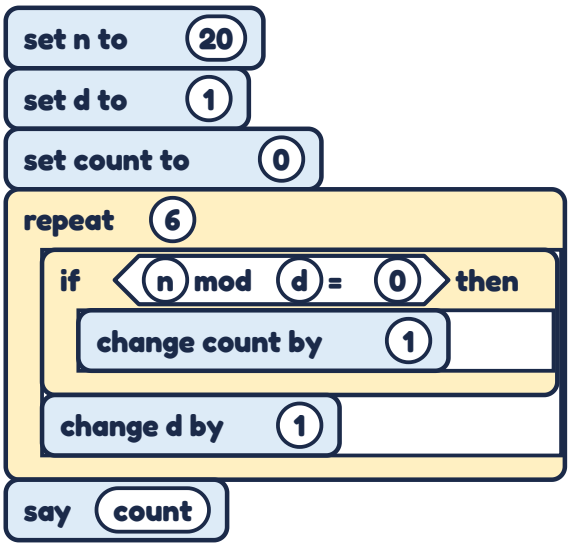
(c) The last number it says is ☐.



Start here

Nearly there. You can do it.

9 **This code counts the factors of n. Fill in the trace table.**



d	n mod d	count
1		
2		
3		
4		
5		
6		

(a) The code says ☐.

(b) To count all the factors of 20, change repeat 6 to repeat ☐.

10 **Most numbers have an even number of factors. Some do not.**

(a) 81 has ☐ factors.

(b) The factor of 81 that pairs with itself is ☐.

(c) Which has an odd number of factors? 50 64 72

11 **Ali has 36 cards. He puts them in equal rows, with at least 2 rows and at least 2 cards in each row.**

(a) The number of rows could be

_____

(b) There are ☐ different ways.

Think harder

Hints are upside down. Try first!

1 **Find the smallest number with exactly this many factors.**

(a) 5 factors: ☐ **(b)** 9 factors: ☐ **(c)** 10 factors: ☐

Working: _____

_____

Hint. Count the factors of 12, 16, 24, 36 and 48. Look for a match.

2 **A number is 2 × 3 × p, where p is a prime greater than 5. The number is less than 100.**

(a) The possible numbers are _____

(b) Each of them has ☐ factors.

(c) Which of them is a multiple of 11? 42 66 78

Working: _____

_____

Hint. Try p = 7, then the next primes. Stop at 100.

3 **Find all the primes between 100 and 120.**

(a) The primes are _____

(b) Pairs of primes that differ by 2: ☐ pairs.

(c) Two odd numbers between 100 and 120 are multiples of 7. They are ☐ and ☐.

Working: _____

_____

Hint. Cross out even numbers first. Then test 3, 5 and 7.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 Beyond Level 6: Counting factors.

Break a number into primes, then count its factors. Each factor uses some of the primes.

(a) List the factors of 72, then count them: ☐

(b) 180 = 2 × 2 × 3 × 3 × 5 has ☐ factors.

(c) The smallest number with exactly 12 factors is ☐.

Factors of 72: _____

Hint. A factor uses 0 to 3 twos and 0 to 2 threes. Count choices.

2 Use the sieve to find the primes up to 60. Start by crossing out 1. Then do this for 2, 3, 5 and 7: ring the number and cross out its other multiples.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60

(a) There are ☐ primes up to 60. (b) The 10th prime is ☐.

Hint. Why can you stop at 7? Think of 8 × 8.

3 This code looks for the smallest factor of n that is greater than 1. Run it.

(a) With n = 221 it says ☐.

(b) With n = 221 the repeat runs ☐ times.

(c) With n = 97 it says ☐.

```
set d to 2
repeat until (n mod d = 0)
    change d by 1
say d
```

Hint. Write d each time. It stops when n mod d is 0.

✔ Check yourself

No hints. Try each one on your own.

1 **List all the factors of 40.** _____

Working: _____

2 **Is 147 prime?** **Yes** **No**

Working: _____

3 **Write 140 as a product of primes.**

_____

Working: _____

4 **47 mod 6 =** ☐ **and 100 mod 8 =** ☐ **.**

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Code the pattern	☐	☐	☐	☐	/ 4	
2	Prime hunters	☐	☐	☐	☐	/ 4	
3	Below zero	☐	☐	☐	☐	/ 4	
4	Algebra machines	☐	☐	☐	☐	/ 4	
5	Every possible way	☐	☐	☐	☐	/ 4	
6	Fair or unfair?	☐	☐	☐	☐	/ 4	
7	Turtle geometry	☐	☐	☐	☐	/ 4	
8	Secret codes	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Showcase challenge	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 6.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 1 · Code the pattern

Your turn
page 6

Answer: 7, 4, 696
Why: The difference is 7, so compare with 7, 14, 21, 28. Each term is 4 less than the number under it. The rule is 7n − 4. The 100th term is 7 × 100 − 4 = 696.

START HERE

1
page 7

Answer: (a) 17, 5 (b) 5, 2 (c) 202 (d) 18
Why: Count: pattern 1 has 7 sticks, pattern 2 has 12, pattern 3 has 17. Each new column adds 5 sticks, so the rule is 5n + 2 (5 × 1 + 2 = 7). Pattern 40: 5 × 40 + 2 = 202. For 92 sticks: 5n + 2 = 92, so 5n = 90 and n = 18.

2
page 7

Answer: (a) 6, 6n + 2, 602 (b) 5, 5n − 2, 498
Why: (a) The difference is 6. Compare with 6, 12, 18, 24: each term is 2 more, so 6n + 2 and 6 × 100 + 2 = 602. (b) The difference is 5. Each term is 2 less than 5, 10, 15, 20, so 5n − 2 and 5 × 100 − 2 = 498.

3
page 7

Answer: (a) 21 (b) 297
Why: Loop 1: it says 3, then x becomes 9. Loop 2: says 9, x becomes 15. Loop 3: says 15, x becomes 21. Loop 4: says 21, x becomes 27. The last number said is 21 but x ends as 27. With repeat 50 it says the 1st term, then 49 jumps later it says 3 + 49 × 6 = 297.

4
page 8

Answer: (a) Yes, term 31 (b) No (c) 302
Why: (a) 157 − 2 = 155 = 5 × 31, so 157 is the 31st term. (b) 205 − 2 = 203 is not a multiple of 5. The terms end in 2 or 7, so 205 is not one. (c) 5 × 60 + 2 = 302.

5
page 8

Answer: (a) 11 (b) 15 (c) 7 (d) 109
Why: The 1st term is 11, so set n to 11. Each jump adds 7. The sprite says a term, then changes n, and 15 terms need 15 says: repeat 15. The last number said is the 15th term: 7 × 15 + 4 = 109.

6
page 8

Answer: (a) 154 (b) 149 (c) 29
Why: The loop adds 5 thirty times: 4 + 5 × 30 = 154. That is the 31st term. The 30th term is 5 × 30 − 1 = 149. From the 1st to the 30th term there are only 29 jumps, so use repeat 29.

7
page 8

Answer: (a) 5n − 1 (b) 3n + 7 (c) 8n − 5
Why: Put n = 1 and n = 2 into each rule. (a) 5 × 1 − 1 = 4 and 5 × 2 − 1 = 9. (b) 3 × 1 + 7 = 10 and 3 × 2 + 7 = 13. (c) 8 × 1 − 5 = 3 and 8 × 2 − 5 = 11. The other rules fail on the 1st term.

8
page 9

Answer: (a) 36 (b) 7, 8 (c) 218 (d) 28
Why: Each day adds $7, so the rule is 7n + 8 (7 × 1 + 8 = 15). (a) Day 4: 7 × 4 + 8 = 36. (c) Day 30: 7 × 30 + 8 = 218. (d) 7n + 8 is more than 200 when 7n is more than 192. 7 × 27 = 189 is too small, 7 × 28 = 196 works: Day 28 (he has $204).

9
page 9

Answer: (a) 59 (b) 41 (c) 17
Why: The 1st term needs 0 repeats and each repeat adds 1 to the term number. (a) 60 − 1 = 59 repeats. (b) 1 + 40 = the 41st term. (c) 25 − 8 = 17 repeats. The repeats count the jumps, not the terms.

10
page 9

Answer: (a) 4, 7, 10 (b) 25 (c) 4, 1
Why: (a) n = 1, 2, 3 give 3 × 1 + 1 = 4, 7, 10. (b) The 8th number is 3 × 8 + 1 = 25. (c) 5, 9, 13 go up by 4, so use 4 × n + 1: 4 × 1 + 1 = 5.

11
page 9

Answer: (a) No (b) 6, 2 (c) 6, 4
Why: The rule 6n gives 6 for the 1st term, but the 1st term is 2. Each term is 4 less than 6n: 6 − 4 = 2, 12 − 4 = 8. The difference only tells you the times table. You must still compare with the terms.

THINK HARDER

1
page 10

Answer: (a) 63 (b) 13 (c) 12
Why: (a) The terms go down by 5, so take 5n from a number. 5 × 1 = 5 and 58 + 5 = 63, so the rule is 63 − 5n. (b) 63 − 50 = 13. (c) 63 − 5n stays above 0 while 5n is less than 63. 5 × 12 = 60 gives 3 and 5 × 13 = 65 would be below 0, so term 12.

2
page 10

Answer: (a) 21 (b) 20 (c) 86
Why: The 5th term is 4 × 5 + 2 = 22. Q adds 4 sixteen times: 22 + 64 = 86. Q has moved 16 terms on, so it ends at term 5 + 16 = 21. P starts at the 1st term, so it needs 21 − 1 = 20 repeats. Check: 6 + 20 × 4 = 86.

3
page 10

Answer: (a) 5 (b) 6 (c) 5 (d) 6, 1
Why: From the 4th to the 9th term is 9 − 4 = 5 jumps. The terms rose by 30, so each jump is 30 ÷ 5 = 6. Going back 3 jumps from the 4th term: 23 − 18 = 5. The rule is 6n − 1 (6 × 4 − 1 = 23).

GO FURTHER

1
page 11

Answer: (a) 10, 18 (b) 28, 40 (c) 130
Why: The extra sticks each time are 6 (4 to 10), then 8 (10 to 18). They go up by 2 each time, so the next extras are 10 and 12: 18 + 10 = 28 and 28 + 12 = 40. For staircase 10 add the extras 6, 8, ..., 22 to 4: 4 + 126 = 130. (The rule is n × n + 3n, which is not a times table plus a number.)

2
page 11

Answer: (a) 4, 5 (b) 31 (c) 65
Why: Places 1, 3, 5, ... say 3, 7, 11, 15: they go up by 4. Places 2, 4, 6, ... say 20, 25, 30, 35: they go up by 5. Place 15 is the 8th of the first kind: 3 + 7 × 4 = 31. Place 20 is the 10th of the second kind: 20 + 9 × 5 = 65.

3
page 11

Answer: (a) 14 (b) 105
Why: t after each repeat: 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105. The repeat stops the first time t is above 100, which is after repeat 14, so k is 14 and t is 105. (These are the triangle numbers 1, 3, 6, 10, ...: the amount added is 1 more each time.)

CHECK YOURSELF

1 page 12
Answer: 7, 2, 702
Why: The difference is 7. Each term is 2 more than 7, 14, 21, 28, so the rule is 7n + 2. The 100th term is 7 × 100 + 2 = 702.

2 page 12
Answer: No
Why: The rule is 5n + 2. 125 – 2 = 123 is not a multiple of 5. The terms end in 2 or 7, so 125 is not a term.

3 page 12
Answer: 128, term 21
Why: 8 + 20 × 6 = 128. From the 1st term to the 21st term there are 20 jumps, so 128 is the 21st term. Check: 6 × 21 + 2 = 128.

4 page 12
Answer: 8
Why: The code changes n before it says it. 5 + 3 = 8, so the first number said is 8, not the 1st term 5. To say 5 first, say n before changing it.

Unit 2 · Prime hunters

Your turn page 13
Answer: No, 11 × 13
Why: Test 2, 3, 5 and 7: each leaves a remainder (143 mod 7 = 3). Then test 11, because 11 × 11 = 121 is less than 143: 11 × 13 = 143. So 143 is not prime.

START HERE

1 page 14
Answer: (a) 1, 2, 3, 6, 7, 14, 21, 42 (b) 9 (c) 10
Why: 42: 1 × 42, 2 × 21, 3 × 14, 6 × 7, so 8 factors. 100: 1 × 100, 2 × 50, 4 × 25, 5 × 20 and 10 × 10. The pairs give 8 factors and 10 pairs with itself, so 9 in all.

2 page 14
Answer: 53, 59, 67, 73, 79
Why: Test each number. 51 = 3 × 17, 57 = 3 × 19, 63 = 7 × 9, 69 = 3 × 23 and 77 = 7 × 11 are not prime. The primes are 53, 59, 67, 73 and 79. They have no factor except 1 and themselves.

3 page 14
Answer: (a) 2 × 2 × 3 × 7 (b) 2 × 3 × 3 × 5
Why: 84 = 4 × 21, 4 = 2 × 2 and 21 = 3 × 7. The ends of the tree are 2, 2, 3 and 7, so 84 = 2 × 2 × 3 × 7. 90 = 9 × 10 = 3 × 3 × 2 × 5, so 90 = 2 × 3 × 3 × 5.

4 page 14
Answer: 3 + 23, 7 + 19, 13 + 13
Why: Try each prime and see if 26 take away that prime is also prime: 26 – 3 = 23, 26 – 7 = 19 and 26 – 13 = 13 all work. 26 – 2, 26 – 5, 26 – 11, 26 – 17 are 24, 21, 15 and 9, which are not prime. There are 3 ways.

5 page 15
Answer: (a) Yes (b) Yes (c) No
Why: (a) 372: the last 2 digits, 72, are in the 4 times table (72 = 4 × 18), so Yes. (b) 2457: the digits add to 2 + 4 + 5 + 7 = 18, a multiple of 3, so Yes. (c) 6835: the digits add to 22, which is not a multiple of 9, so No.

6 page 15

Answer: (a) 1, 2, 3, 1 (b) Yes (c) No, 7 × 17
Why: 113 = 56 × 2 + 1, 113 = 37 × 3 + 2, 113 = 22 × 5 + 3 and 113 = 16 × 7 + 1. No remainder is 0, and the next prime 11 is too big (11 × 11 = 121), so 113 is prime. 119 = 7 × 17, so 119 is not prime.

7 page 15

Answer: (a) 53 (b) 71
Why: (a) The primes between 40 and 60 are 41, 43, 47, 53 and 59. Only 53 has digits that add to 8. (b) The primes between 60 and 80 are 61, 67, 71, 73 and 79. Adding 2 gives 63, 69, 73, 75, 81, and only 73 is prime, so I am 71.

8 page 15

Answer: (a) 14 (b) 5 (c) 70
Why: x goes 7, 14, 21, 28, 35, 42, 49, 56, 63, 70. The code says x only when x mod 2 = 0, that is when x is even. It says 14, 28, 42, 56 and 70. The first is 14, there are 5 numbers, and the last is 70.

9 page 16

Answer: (a) 4 (b) 20
Why: n mod d for d = 1 to 6: 0, 0, 2, 0, 0, 2, so the count goes 1, 2, 2, 3, 4, 4. The code says 4. But 20 has 6 factors: 1, 2, 4, 5, 10 and 20. The loop stops at d = 6 and misses 10 and 20. To count every factor, repeat 20 times.

10 page 16

Answer: (a) 5 (b) 9 (c) 64
Why: (a) 81: 1 × 81, 3 × 27 and 9 × 9 give 1, 3, 9, 27, 81. (b) 9 is paired with itself. (c) 64 = 8 × 8 is a square, so it has an odd number (7). 50 has 6 factors and 72 has 12.

11 page 16

Answer: (a) 2, 3, 4, 6, 9, 12, 18 (b) 7
Why: The number of rows must be a factor of 36 other than 1 and 36: 2, 3, 4, 6, 9, 12 and 18. That is 7 ways. (2 rows of 18 and 18 rows of 2 count as different ways.)

THINK HARDER

1 page 17

Answer: (a) 16 (b) 36 (c) 48
Why: (a) 16 has 1, 2, 4, 8, 16: 5 factors, and no smaller number has 5. (b) 36 has 1, 2, 3, 4, 6, 9, 12, 18, 36: 9 factors. (c) 48 has 1, 2, 3, 4, 6, 8, 12, 16, 24, 48: 10 factors. Numbers below 48 have fewer.

2 page 17

Answer: (a) 42, 66, 78 (b) 8 (c) 66
Why: p can be 7, 11 or 13: 2 × 3 × 7 = 42, 2 × 3 × 11 = 66 and 2 × 3 × 13 = 78. p = 17 gives 102, which is too big. 42 has factors 1, 2, 3, 6, 7, 14, 21, 42, so 8 factors. 66 and 78 also have 8.

3 page 17

Answer: (a) 101, 103, 107, 109, 113 (b) 2 (c) 105 and 119
Why: Test 2, 3, 5 and 7 only, because 11 × 11 = 121 is above 120. 101, 103, 107, 109 and 113 have no factor from the list. The odd non-primes are 105 (5), 111 (3), 115 (5), 117 (3) and 119 (7). Pairs that differ by 2: 101 and 103, 107 and 109. (c) The odd multiples of 7 are 105 = 7 × 15 and 119 = 7 × 17.

GO FURTHER

1
page 18

Answer: (a) 12 (b) 18 (c) 60
Why: (a) 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72 is 12 factors. In 2 × 2 × 2 × 3 × 3 a factor uses 0 to 3 twos (4 choices) and 0 to 2 threes (3 choices): 4 × 3 = 12. (b) 180 = 2 × 2 × 3 × 3 × 5: 3 × 3 × 2 = 18 factors. (c) 60 = 2 × 2 × 3 × 5 has 3 × 2 × 2 = 12 factors. Smaller numbers have fewer.

2
page 18

Answer: (a) 17 (b) 29
Why: Cross out multiples of 2, 3, 5 and 7. The next prime, 11, has 11 × 11 = 121, which is above 60, so you can stop at 7. The numbers left are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59: that is 17 primes. The 10th is 29.

3
page 18

Answer: (a) 13 (b) 11 (c) 97
Why: n = 221: d goes 2, 3, … and stops at 13, because 221 mod 13 = 0. It says 13, after 11 repeats (d = 2 to 13). n = 97: d keeps going until d = 97, because 97 is prime. It says 97. If the code says n itself, then n is prime.

CHECK YOURSELF

1
page 19

Answer: 1, 2, 4, 5, 8, 10, 20, 40
Why: 40 = 1 × 40 = 2 × 20 = 4 × 10 = 5 × 8. The factors are 1, 2, 4, 5, 8, 10, 20 and 40.

2
page 19

Answer: No
Why: The digits add to 1 + 4 + 7 = 12, a multiple of 3, so 3 divides 147 (147 = 3 × 49). It is not prime.

3
page 19

Answer: 2 × 2 × 5 × 7
Why: 140 = 2 × 70 = 2 × 2 × 35 = 2 × 2 × 5 × 7. Every number here is prime.

4
page 19

Answer: 5, 4
Why: 47 = 7 × 6 + 5, so the remainder is 5. 100 = 12 × 8 + 4, so the remainder is 4.